In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – II
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**AC**	**ABC**	**ACD**
2.	**ACD**	**ABC**	**ABC**
3.	**BD**	**B**	**AB**
4.	**ABC**	**ABC**	**ACD**
5.	**AC**	**ABCD**	**A**
6.	**ABCD**	**AB**	**AB**
7.	**ACD**	**AC**	**ABCD**
8.	**ABCD**	**AB**	**ABC**
9.	**AC**	**AC**	**AD**
10.	**AC**	**AB**	**AB**
1.	**(A) → r,s** **(B) → r,s** **(C) → r,s** **(D) → r,t**	**(A) → p,q** **(B) → q,r** **(C) → r,s** **(D) → p,s**	**(A) → s** **(B) → r** **(C) → q** **(D) → p**
2.	**(A) → p** **(B) → q** **(C) → s** **(D) → r**	**(A) → p,q,r** **(B) → r** **(C) → q,s** **(D) → q,r**	**(A) → p** **(B) → p,q** **(C) → p** **(D) → r,s**
1.	**1**	**6**	**5**
2.	**3**	**4**	**3**
3.	**1**	**6**	**2**
4.	**6**	**4**	**6**
5.	**1**	**3**	**2**
6.	2	**5**	**3**
7.	**5**	**7**	**4**
8.	**5**	**1**	**6**

Physics

PART – I

SECTION – A

1. AC

Sol. Mass per unit length of the wire = 0.1 kg/m.

At any instant, tension in the string is (2 + 2.25 t) g N

$$v = \sqrt{\frac{(2+2.25t)g}{0.1}} \quad ; \quad \int_0^{10} dx = 10\int_0^t \sqrt{2+2.25t}\, dt$$

t = 0.612 s

$$T = (2 + 2.25 \times 0.612) \times 10 = 33.77 \text{ N}$$

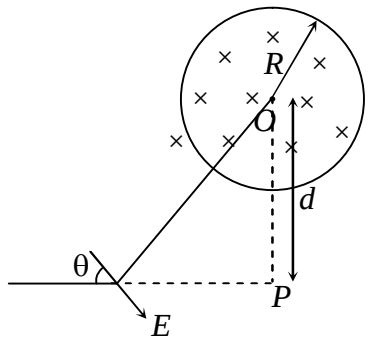
2. ACD

Sol. $\int \vec{E} \cdot d\vec{l} = A\frac{dB}{dt}$

$$E2\pi\sqrt{x^2+d^2} = \pi R^2 k$$

$$E = \frac{\pi R^2 k}{2\sqrt{x^2+d^2}}$$

$$W_{ext} = \int_0^\infty q\vec{E} \cdot dx = \frac{q\pi R^2}{4}k$$



3. BD

Sol. Length $\propto G^x c^y h^z$

$$L = \left[M^{-1}L^3T^{-2}\right]^x \left[LT^{-1}\right]^y \left[ML^2T^{-1}\right]^Z$$

By comparing the power of M, L and T in both sides we get

$-x + z = 0, 3x + y + 2z = 1$ and $-2x - y - z = 0$

By solving above three equations we get

$$x = \frac{1}{2}, y = -\frac{3}{2}, z = \frac{1}{2}$$

4. ABC

Sol. Assume x-axis along OA and y-axis perpendicular to it in the plane of ring.

After 1st collision

Velocity of particle = $\frac{v}{\sqrt{2}}$ along x-axis

And velocity of ring = $\frac{v}{\sqrt{2}}$ along y-axis

After second collision velocity of ring = $\frac{v}{\sqrt{2}}\hat{i} + \frac{v}{\sqrt{2}}\hat{j}$

Velocity of particle = 0

Time to return at point A = $4 \times \frac{R\sqrt{2}}{v}$

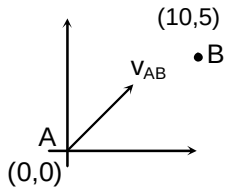
5. AC

Sol. $\vec{v}_{AB} = (3-a)\hat{i} + (3-b)\hat{j}$

$\vec{a}_{AB} = \vec{o}$

$(3-a)\times 2 = 10$ and $(3-b)\times 2 = 5$

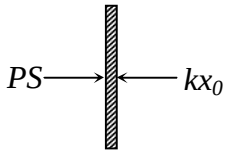
a = –2 and b = $\frac{1}{2}$



6. ABCD

Sol. Equilibrium of piston gives

$PS = kx_0$ or $P = \frac{kx_0}{S}$



Since, the chamber is thermally insulated $\Delta Q = 0$

$\therefore$ Elastic potential energy of spring = work done by gas

or, work done by gas = $\frac{1}{2}kx_0^2$

This work is done in the expense of internal energy of the gas.

Therefore, internal energy of the gas is decreased by $\frac{1}{2}kx_0^2$.

Internal energy of an ideal gas depends on its temperature only. Internal energy of the gas is decreasing. Therefore, temperature of the gas will decrease.

7. ACD

Sol. $\vec{F}_m = q(\vec{v}\times\vec{B})$ and apply Newton's second law of motion.

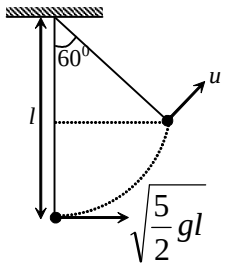
8. ABCD

Sol. $T - mg\cos 60^0 = \frac{mv^2}{l}$ … (i)

$\frac{1}{2}m\left(\sqrt{\frac{5}{2}gl}\right)^2 - mg\frac{l}{2} = \frac{1}{2}mv^2$ … (ii)

$v = \sqrt{\frac{3}{2}gl}$ and $T = 2mg$

Net force is in horizontal direction when $\theta = 60^0$



9. AC

Sol. Equivalent resistance across capacitor (after shorting the battery) = $\frac{11R}{4}$.

Voltage across capacitor in steady state = $\frac{3E}{4}$.

$q = \frac{CE}{4}\left(1 - e^{\frac{-4t}{11RC}}\right)$

10. AC

Sol. $\tau = \dfrac{L}{Req} = \dfrac{2}{8} = \dfrac{1}{4}$

at steady state inductor short circuited

Hence i = 0.75

SECTION – B

1. $A \to rs$ $B \to rs$ $C \to rs$ $D \to rt$

Sol. For cases (a), (b) and (c), magnetic field due to I_1 and I_2 are mutually perpendicular. Therefore, they do not exert any force on each other.
For case (d), magnetic fields at point P due to I_1 and I_2 are in the same direction.

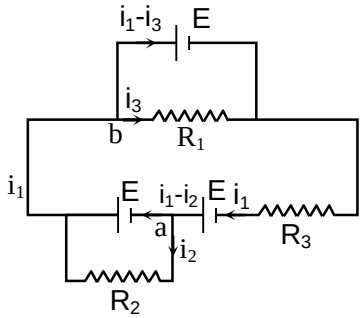
2. $A \to p$ $B \to q$ $C \to s$ $D \to r$

Sol. On solving by KVL we get, $i_1 = \dfrac{E}{R}$, $i_2 = -\dfrac{E}{R}$, $i_3 = \dfrac{E}{R}$

$\Rightarrow i_1 - i_2 = \dfrac{2E}{R}$ and $i_1 - i_3 = 0$

So reading of ammeter A_1 is $E/R = 4/3$

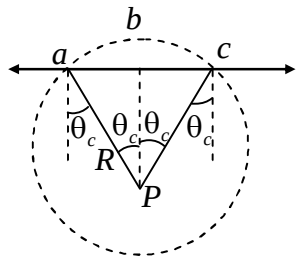
A_2 is $2E/R = 8/3$, A_3 is zero and $\Delta V_{ab} = 4V$



SECTION – C

1. 1

Sol. The light escape is confined within a cone of apex angle '$2\theta_c$' where θ_c is the critical angle. Imagine a sphere with source of light as its centre and the surface area abc is A.



here $A = \int_0^{\theta_c} 2\pi R^2 \sin\theta \, d\theta = 2\pi R^2(1 - \cos\theta_c)$

$= \pi R^2 \qquad \left[\therefore \theta_c = \sin^{-1}\left(\dfrac{\sqrt{3}}{2}\right) = 60^o\right]$

$\therefore$ Power transfer $= P \times \dfrac{A}{4\pi R^2} = 4 \times \dfrac{1}{4} = 1W$

2. 3

Sol. Applying lens formula, $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$

$-\dfrac{1}{v^2}\dfrac{dv}{dt} - \dfrac{1}{u^2}\dfrac{du}{dt} = 0$

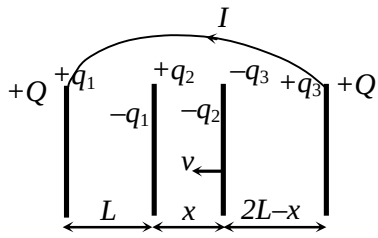
$\dfrac{dv}{dt} = v_0$

$\frac{dv}{dt} = v_1 + v_0$ (v_1 = velocity of object towards lens)

$\therefore \quad v_1 + v_0 = \frac{1}{n^2} v_0, \quad v_1 = \left(\frac{1-n^2}{n^2}\right) v_0, \quad v_1 = 3v_0 = 3$ m/s

3. 1

Sol. $\frac{2q_1 L}{2A\varepsilon_0} + \frac{2q_2 x}{2A\varepsilon_0} - \frac{2q_3(2L-x)}{2A\varepsilon_0} = 0$ … (i)



$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt}$ … (ii)

From (i),

$q_1 L + q_2 x - 2q_3 L + q_3 x = 0$

$\Rightarrow \quad L\frac{dq_1}{dt} + x\frac{dq_2}{dt} + q_2\frac{dx}{dt} - 2\frac{dq_3}{dt}L + x\frac{dq_3}{dt} + q_3\frac{dx}{dt} = 0$

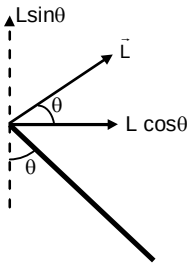
$\Rightarrow \quad I = \frac{v}{3L}(q_2 + q_3) = 1$ mA

4. 6

Sol. $\left|\frac{d\vec{L}}{dt}\right| = L\cos\theta\omega$

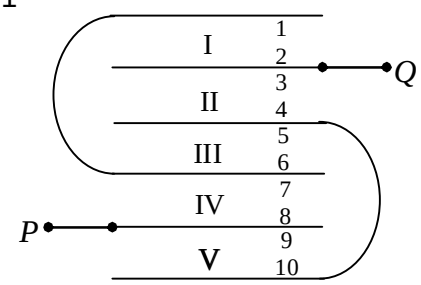
$= \frac{m\ell^2}{3}\omega\sin\theta\cos\theta\omega$

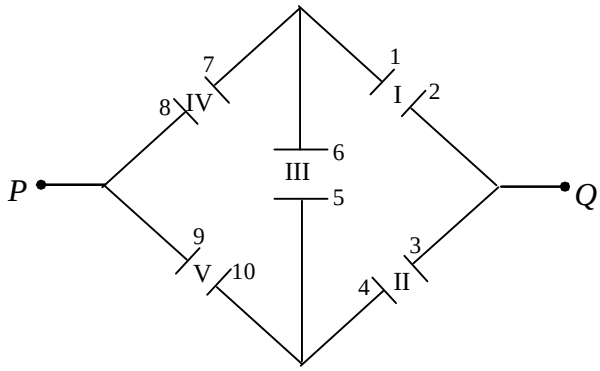
$= \frac{m\ell^2}{3}\omega^2\sin\theta\cos\theta$



5. 1

Sol.

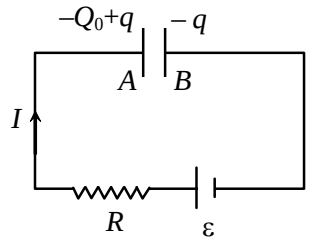




Equivalent capacity, $C_{PQ} = C = \frac{\varepsilon_0 A}{d}$

6. 2

Sol. Let at any time t charge flown through the plate B to plate A is q and instantaneous current is I.



From loop theorem $\left(\frac{2q-Q_0}{2C}\right)+IR-\varepsilon=0$

$\Rightarrow \quad R\frac{dq}{dt}=\frac{-2q+2\varepsilon C+Q_0}{2C}$

$\Rightarrow \quad \frac{dq}{2\varepsilon C+Q_0-2q}=\frac{dt}{2RC}$

Now for charge on plate *A* to be zero $q = Q_0$.

Integrating $\int_0^{Q_0}\frac{dq}{2\varepsilon C+Q_0-2q}=\int_0^t\frac{dt}{2RC}$

$= \quad t=RC\ln\left[\frac{2\varepsilon C+Q_0}{2\varepsilon C-Q_0}\right]$

Putting the value of C, Q_0, ε and R
We get $t = 2$ seconds.

7. 5

Sol. K.E. = $\frac{1}{2}mv^2+\frac{1}{2}mv^2+\frac{1}{2}\frac{m(2r)^2}{12}w^2+\frac{1}{2}mr^2w^2$ = 5.

8. 5

Sol. At stable equilibrium, U is minimum.

$\frac{dU}{dx}$ = 0 and $\frac{d^2U}{dx^2}$ > 0

$= \frac{1}{dx}\left(\frac{x^3}{3}-\frac{ax^2}{2}+20x\right)=0.$

$\Rightarrow \quad x^2 - 9x + 20 = 0. \quad \Rightarrow \quad (x - 5)\,(x - 4) = 0.$

x = 5 and x = 4 are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2}$ > 0. i.e. at x = 5.

Chemistry

PART – II

SECTION – A

1. ABC

Sol. $KO \xrightarrow{H_2O} O_2$ gas

$CaC_2 \xrightarrow{H_2O} C_2H_2$ gas

$Mg_3N_2 \xrightarrow{H_2O} NH_3$ gas

$Ba(NO_3)_2 \xrightarrow{H_2O} Ba^{2+} + 2NO_3^-$

2. ABC

Sol. Neither the lone pair of oxygen nor the pi – bonds are in conjugation in (D).

3. B

Sol. $COCl_2 + H_2O \longrightarrow CO_2 + 2HCl$

4. ABC

Sol. Silicones are the polymers of the hydrolysis products of alkyl silicon chlorides.

5. ABCD

Sol. The boiling point of water is 100^oC.

$\therefore$ Dissolution of any non-volatile solute of any quantity in water will cause elevation in boiling point. So, the boiling point of aqueous solution of any non-volatile solute is greater than 100^oC.

6. AB

Sol. In C the net charge on cationic as well as anionic parts of the complex is zero. Such complexes are not formed. D is not an isomer of the complex due to different molecular formula and molecular mass.

7. AC

Sol. (A) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$

(B) $MgCl_2.6H_2O \xrightarrow{\Delta} MgCl_2.H_2O + 5H_2O$

$Na_2SO_4.10H_2O \xrightarrow{\Delta} Na_2SO_4 + 10H_2O$

(C) $2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$

$Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$

8. AB

Sol. All aliphatic aldehydes respond to Fehling test. Benzaldehyde does not give Fehling's test.

9. AC

Sol. The coordination number in both the structure is 12 and the packing fraction is 74%

10. AB

Sol. • The dipole moment of NH_3 is greater than NF_3 because the bond pair dipole moment is enhanced by the lone pair dipole moment in NH_3. In NF_3 the resultant of bond pair dipole moment is opposed by the lone pair dipole moment.

• $CHCl_3$ is unsymmetrical, hence it contains certain dipole moment whereas CCl_4 is symmetrical with zero dipole moment.

SECTION – B

1. $A \rightarrow pq, B \rightarrow qr, C \rightarrow rs, D \rightarrow ps$

Sol. (A) $Fe^{2+} + 2KCN \longrightarrow Fe(CN)_2 \downarrow \xrightarrow[\text{excess}]{4KCN} K_4\left[Fe(CN)_6\right]$

$Fe(OH)_2$ is not soluble in excess NaOH and excess NH_4OH

(B) HgO is not soluble in excess NaOH and excess NH_4OH

$Hg^{2+} + 2KI \longrightarrow HgI_2 \downarrow \xrightarrow{2KI} K_2\left[HgI_4\right]$

(C) $Pb^{2+} + 2KI \longrightarrow PbI_2 \downarrow \xrightarrow{2KI} K_2\left[PbI_4\right]$

$Pb^{2+} + H_2S \longrightarrow PbS \downarrow + 2H^+$

$3PbS + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 2NO + 3S + 4H_2O$

(D) $Ag^+ + KCN \longrightarrow AgCN \downarrow \xrightarrow{KCN} K\left[Ag(CN)_2\right]$

$Ag^+ + H_2S \longrightarrow Ag_2S \downarrow + 2H^+$

$3Ag_2S + 8HNO_3 \longrightarrow 6AgNO_3 + 2NO + 3S + 4H_2O$

2. $A \rightarrow pqr, B \rightarrow r, C \rightarrow qs, D \rightarrow qr$

Sol. (A) $pH = p^{K_a} + \log\frac{[HCOONa]}{[HCOOH]} = 3.74 + \log\frac{0.1}{0.2} = 3.74$

$\therefore$ pH < 7

(B) pH = -log $[H^+]$ = -log10^{-1} = 1, $\therefore$ pH < 7

CH_3COOH cannot dissociated in presence of HCl due to common ion effect.

(C) $CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O$

Hydrolysis of CH_3COONH_4 will take place

$$pH = \frac{1}{2}\left(p^{K_w} + p^{K_a} - p^{K_b}\right) = \frac{1}{2}\left[14 + 4.74 - (14 - 9.26)\right]$$

= ½ [14 + 4.74 – 4.74] = 7. $\therefore$ pH = 7 and the solution is a buffer

(D)

	$HCOONa + HCl \longrightarrow HCOOH + NaCl$			
Initial meq	30	10	0	0
Meq. After reac^n	20	0	10	10

It is a buffer, $pH = p^{K_a} + \log\frac{[HCOONa]}{[HCOOH]} = 3.74 + \log\frac{20}{10} = 4.04, < 7$

SECTION – C

1. 6

Sol. $\Delta H = nC_v\Delta T$

$$\text{or, } 12 = \frac{4.48}{22.4} \times C_V \times 15$$

or, $C_v = 4$

$C_P - C_V = R = 2 \Rightarrow C_P = C_V + 2 = 4 + 2 = 6$ cal

2. 4

Sol. The products are $CH_3CH_2CH_2NO_2$, $CH_3\underset{\underset{NO_2}{|}}{C}HCH_3$, $CH_3CH_2NO_2$ and CH_3NO_2 because C–C and C – H bond cleavage takes place in this reaction.

3. 6

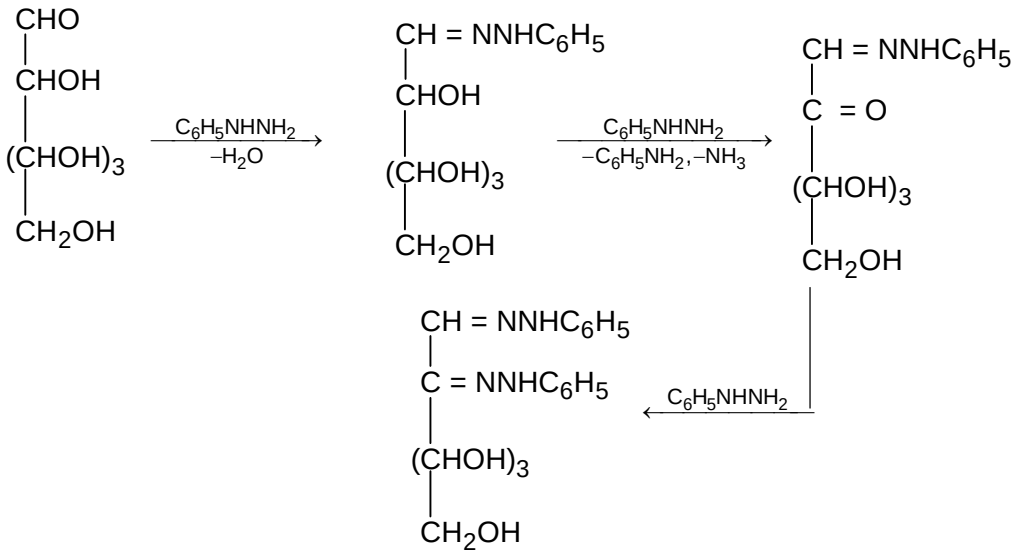
Sol. For zero order reaction the half-life is directly proportional to the concentration of reactant.

$$X \xrightarrow{t_{1/2}=4\,sec} \frac{X}{2} \xrightarrow{t_{1/2}=2\,sec} \frac{X}{4}$$

$\therefore$ Total time needed for 75% reaction = 4 + 2 = 6 sec

4. 4

Sol.



5. 3

Sol.

$$2X(g) \rightleftharpoons 2Y(g) + Z(g)$$

Initial	1	0	0
At equi	$1 - \alpha$	α	$\alpha/2$

$\therefore$ Partial pressure [If total pressure P]

$$p_x = \left[\frac{1-\alpha}{1+\alpha/2}\right]P \simeq P \quad p_y = \frac{\alpha}{1+\alpha/2}P \simeq \alpha P$$

$$p_z = \left[\frac{\alpha/2}{1+\alpha/2}\right]P \simeq \frac{\alpha P}{2}$$

$$\therefore K_p = \frac{[p_y]^2[p_z]}{[p_x]^2} = \frac{[\alpha P]^2\left[\frac{\alpha P}{2}\right]}{P^2}$$

$$K_p = \frac{\alpha^3 P}{2}$$

$$\alpha = \left(\frac{2K_p}{P}\right)^{\frac{1}{3}}$$

6. 5

Sol. Work done in anticlockwise cycle is positive and in clockwise cycle is negative.
moles of A are present. Total number of moles of A,B and C at the end of 75% reaction are X. What will be the value of 2X

7. 7

Sol. One mole of A will form 2 moles of B and C after completion of reaction when 75% of A converted into B and C then total No of moles X = 0.5 + (2 $\times$ 0.5 + 1$\times$2) = 3.5
Hence 2X = 3.5$\times$2 = 7

8. 1

Sol. $\underset{(0.01-x)}{Sb_2S_3(s)} + \underset{(0.01-3x)}{3H_2(g)} \rightarrow \underset{2x}{2Sb(s)} + \underset{3x}{3H_2S(g)}$

No.of moles of PbS formed = $\frac{1.19}{239} = 0.005$

At eqn. $\left[H^+\right] = \left(\frac{0.005}{250}\right)$

$$K_C = \left(\frac{0.005}{0.005}\right)^3 = 1$$

Mathematics

PART – III

SECTION – A

1. ACD

Sol. On expanding we get $\left(\vec{a}.\vec{c}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\left(\vec{c}\right)=-\vec{b}$

$\Rightarrow \vec{c}-\left(\vec{a}.\vec{c}\right)\vec{a}=\vec{b}$

$\Rightarrow$ (a) is correct and (b) is ruled out.

$\Rightarrow \left|\vec{c}-\left(\vec{a}.\vec{c}\right)\vec{a}\right|^2=\left|\vec{b}\right|^2$

If θ is the angle between $\vec{a}$ and $\vec{c}$ then

$4+\left(2\cos\theta\right)^2-2\left(2\cos\theta\right)^2=1$

$\Rightarrow \cos\theta=\pm\frac{\sqrt{3}}{2} \qquad \Rightarrow \theta=30^o;150^o$

$\Rightarrow$ (C) and (D) is correct.

2. ABC

Sol. $8\sin x=\frac{\sqrt{3}}{\cos x}+\frac{1}{\sin x}$

$\Rightarrow 8\sin^2 x\cos x=\sqrt{3}\sin x+\cos x$

$4\left(1-\cos 2x\right)\cos x=\sqrt{3}\sin x+\cos x$

$3\cos x-2\left(\cos 3x+\cos x\right)=\sqrt{3}\sin x$

$\cos x-2\cos 3x=\sqrt{3}\sin x$

$\frac{1}{2}\cos x-\frac{\sqrt{3}}{2}\sin x=\cos 3x$

$\cos\left(x+\frac{\pi}{3}\right)=\cos 3x$

$3x=2nx\pm\left(x+\frac{\pi}{3}\right)$

(+)	(–)
$3x=2x\pi+x+\frac{\pi}{3}$	$3x=2n\pi-x-\frac{\pi}{3}$
$2x=2n\pi+\frac{\pi}{3}$	$4x=2n\pi-\frac{\pi}{3}$
$x=nx+\frac{\pi}{6}$	$x=\frac{n\pi}{2}-\frac{\pi}{12}$
$[0,2\pi]$	
$\frac{\pi}{6}$	$\frac{\pi}{2}-\frac{\pi}{12}=\frac{6\pi-\pi}{12}=\frac{5\pi}{12}$

$\frac{5\pi}{6}$ $\qquad$ $\pi-\frac{\pi}{12}=\frac{11\pi}{12}$

$\frac{3\pi}{2}-\frac{\pi}{12}=\frac{18\pi-\pi}{12}=\frac{17\pi}{12}$

$2\pi-\frac{\pi}{12}=\frac{23\pi}{12}$

3. AB

Sol. $\det(A+B)=-\det(A)\det(B)\det(A+B)$

$=-\det(A^T)\det(B^T)\det(A+B)$

$=-\det(A^T(A+B)B^T)$

$=-\det(A^T(A+B)B^T)$

$=-\det(A^TAB^T+A^TBB^T)$

$=-\det(B^T+A^T)$

$=-\det(A+B)^T$

$=-\det(A+B)\Rightarrow\det(A+B)=0$

4. ACD

Sol. $g'(0)=\lim_{h\to\infty}\frac{g(0+h)-g(0)}{h}=0$

$\Rightarrow g(x)$ is differentiable $\forall x\in R$

$$g'(x)=\begin{cases}2x\sin\left(\frac{\pi}{x}\right)-\pi\sin\left(\frac{\pi}{x}\right)+2(x-1)\sin\left(\frac{\pi}{x-1}\right)-\pi\sin\left(\frac{\pi}{x-1}\right); & x\neq 1\\ 0\ ; & x=0,1\end{cases}$$

But $\lim_{x\to0}g'(x)=$ does not exist $\neq g'(0)\Rightarrow g'(x)$ is discontinuous at x = 0

Similarly $\lim_{x\to1}g'(x)=$ does not exist.

5. A

Sol. $\lim_{t\to0}\frac{\int_0^t\sin t^2\,dt}{\frac{1}{2}t\sin t^2}=\lim_{t\to0}\frac{2\sin t^2}{\sin t^2+2t^2\cos t^2}=\lim_{t\to0}\frac{2}{1+2\frac{t^2}{\tan t^2}}=\frac{2}{1+2}=\frac{2}{3}$

6. AB

Sol. The point of intersection of diagonal is (0, 0) and one vertex A (2, 1)

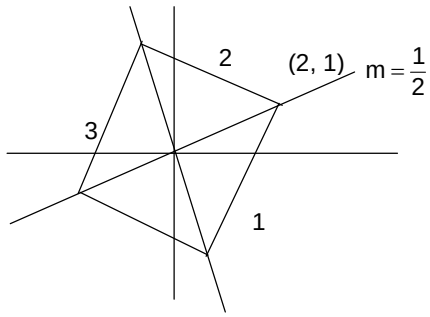
$\Rightarrow$ Vertex is (–2, –1) verties B and D are $\pm i(2+i)$

$=-1+2i,\ 1-2i$

$\Rightarrow B(-1,2)$ and $D(1,-2)$

$\Rightarrow$ Equation of the sides are

$x+3y=\pm5,\ 3x-y=\pm5$.



7. ABCD

Sol. The three planes intersect in a line.

8. ABC

Sol. $(2x-3y+z)^{10}$

General term $=\sum_{\alpha+\beta+\gamma=10}\frac{10!(2x)^{\alpha}(-3y)^{\beta}(z)^{\gamma}}{\alpha!\beta!\gamma!}$ .....(1)

$\therefore$ Number of terms $=\frac{12!}{10!2!}=\frac{12\times11}{2}=66$

Put $x=y=z=1$ in (1), we get

Sum of all the coefficients 0.

Put $\alpha=2,\beta=3,\gamma=5$ in (1),

We get coefficient of $x^2y^3z^5=\frac{10!}{2!.3!.5!}(2)^2(-3)^3=\frac{-10!\times9}{5!}$

9. AD

Sol. Clearly, $\sum_{k=0}^{100}{}^{100}C_k(x-2)^{100-k}3^k=((x-2)+3)^{100}=(1+x)^{100}$

Hence, coefficient of x^{50} is ${}^{100}C_{50}$. Now verify

Option (A), (D) equal to ${}^{100}C_{50}$

(B) $99!\left({}^{100}C_{50}\right)$

(C) ${}^{100}C_{101}$

10. AB

Sol. $\log_3|2z+1|<\log_3|2z-1|$ (as $\cot^{-1}x$ is a decreasing function)

$\therefore\ |2z+1|<|2z-1|$

Or $(2z+1)(2\bar{z}+1)<(2z-1)(2\bar{z}-1)$

$2(z+\bar{z})<-2(z+\bar{z})$

$z+\bar{z}<0$

$2\mathrm{Re}(z)<0$

$\therefore \mathrm{Re}(z)<0$

SECTION – B

1. A $\to$s B $\to$r C $\to$q D $\to$p

Sol. (A) $\{\cos(2A+\theta)+\cos(2B+\theta)\}$

$=2\cos(A-B)\cos(A+B+\theta)$

Maximum value is $|2\cos(A-B)|$ when $\cos(A+B+\theta)=\pm1$

(B) $\{\cos2A+\cos2B\}=2\cos(A+B)\cos(A-B)$

Maximum value is $2\cos(A+B)$ when $\cos(A-B)=1$

(C) For $y=\sec x, x\in\left(0,\frac{\pi}{2}\right)$, tangent drawn to it any point lies completely below the graph of $y=\sec x$, thus $\frac{\sec 2A+\sec 2B}{2}\geq\sec(A+B)$

$\Rightarrow \quad \sec 2A+\sec 2B\geq 2\sec(A+B)$

Hence, minimum value is $A+B$

(D) $\sqrt{\{\tan\theta+\cot\theta-2\cos 2(A+B)\}}$

$=\sqrt{\left(\sqrt{\tan\theta}-\sqrt{\cot\theta}\right)^2+2-2\cos 2(A+B)}$

$=\sqrt{\left(\sqrt{\tan\theta}-\sqrt{\cot\theta}\right)^2+4\sin^2(A+B)}$

Minimum value occurs when $\sqrt{\tan\theta}=\sqrt{\cot\theta}$ and minimum value is $\sqrt{4\sin^2(A+B)}=2\sin(A+B)$

2. $A\rightarrow p$ $B\rightarrow pq$ $C\rightarrow p$ $D\rightarrow rs$

Sol. (A) Equation of the plane π is $A(x-1)+B(y-1)+C(z-1)=0$ where $A+0.B-1.C=0$ and $-A+B+0.C=0\Rightarrow A=B=C$ and the equation of π is $x+y+z=3$. which meets the co – ordinates axes at A (3, 0, 0), B (0, 3, 0) and C (0, 0, 3). So volume of the tetrhadron. $OABC=\frac{1}{6}\begin{vmatrix}0&0&0&1\\3&0&0&1\\0&3&0&1\\0&0&3&1\end{vmatrix}=\frac{1}{6}\times 27=\frac{9}{2}$

(B) $AB=BC=CA=3\sqrt{2}$, so the area of the face ABC which is an equilateral triangle $=\frac{\sqrt{3}}{4}\times\left(3\sqrt{2}\right)^2=\frac{9\sqrt{3}}{2}$. Each of the face OAB, OBC, OCA is an isosceles triangle with sides $3,3,3\sqrt{2}$

So, area of each of these faces is $\frac{1}{2}\times 3\sqrt{2}\sqrt{9-\frac{9}{2}}=\frac{9}{2}$

(C) Required distance is $\left|\frac{\frac{3\sqrt{3}}{2}+3\sqrt{3}+3-3}{\sqrt{1+1+1}}\right|=\frac{9}{2}$

(D) Distance of O from the face $ABC=\left|\frac{0+0+0-3}{\sqrt{1+1+1}}\right|=\sqrt{3}$ distance of A, B, C from opposite faces OBC, OAC, OAB are equal and is 3.

SECTION – C

1. 5

Sol. Note that $z = 0$ is one root of the given equation

$z^4 + az^3 + (12 + 9i)z^2 + bz = 0$

canceling z, we get $z^3 + az^2 + (12 + 9i)z + b = 0$ ……(1)

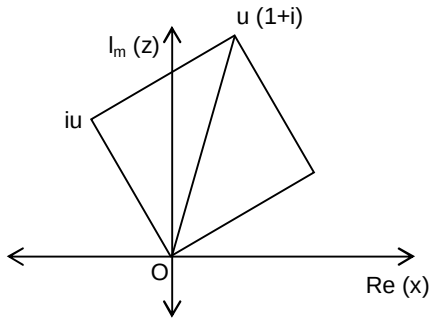
The 3 vertices of the square other than $(0 + 0i)$ are u, iu, u (1 + i)

Now $u(ui) + (iu) + (1+i)u + \left[(1+i)u\right]u = 12 + 9i$ (using theory of equation)

$u^2 = \dfrac{4+3i}{i} = 3 - 4i \Rightarrow u = \sqrt{3-4i} = \pm(2-i)$

$\therefore u = 2 - i$ or $-2 + i$

$\Rightarrow |u|^2 = 5$



2. 3

Sol. $\beta = \displaystyle\int_0^1 \frac{2x + 3x^2 + 4x^3}{2\sqrt{x^4 + x^3 + x^2}}\,dx$

Put $(x^2 + x^3 + x^4) = t \Rightarrow (2x + 3x^2 + 4x^3)dx = dt$

$\beta = \displaystyle\int_0^3 \frac{dt}{2\sqrt{t}}$

$\beta = \left[\sqrt{t}\right]_0^3 = \sqrt{3} \Rightarrow \beta^2 = 3$

3. 2

Sol. Area $= \dfrac{1}{2}ab$

AD : $y = x + 3$

BE = $y = 2x + 4$

solve $G(-1, 2)$

acute angle α between the medians is

$\tan\alpha = \dfrac{m_1 - m_2}{1 + m_1 m_2}$

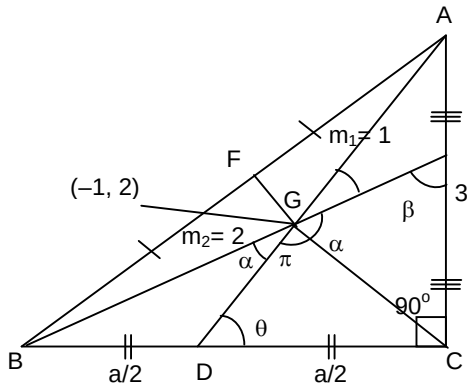
$\tan\alpha = \dfrac{2-1}{1+2} \Rightarrow \tan\alpha = \dfrac{1}{3}$

now, $(180 - \alpha) + 90^o + \theta + \beta = 360^o$

$\Rightarrow \quad \alpha = \theta + \beta - 90^o$

$\cot\alpha = -\tan(\theta + \beta)$

$-3 = \dfrac{\tan\theta + \tan\beta}{1 - \tan\theta\tan\beta}$ or $-3 = \dfrac{\frac{2b}{a} + \frac{2a}{b}}{1 - \frac{2b}{a}.\frac{2a}{b}}$



$\Rightarrow \quad 9 = \dfrac{2\left(a^2 + b^2\right)}{ab}$

$9ab = 2 \times 3600$

$\Rightarrow \quad \dfrac{1}{2}ab = 400$

$\therefore$ Area = 400 sq. units

4. 6

Sol. $\dfrac{dy}{dt} = -k\sqrt{y}$

$\dfrac{dy}{\sqrt{y}} - k\,dt$

$2\sqrt{y} = -kt + c$

$t = 0;\ y = 4$

$4 = c$

$2\sqrt{y} = -kt + 4$

$y = 0$

$kt = 4$

$t = \dfrac{4}{\dfrac{1}{15}} = 60 \text{ minutes}$

5. 2

Sol. $\text{Sum} = \sum_{n=1}^{n} \tan^{-1}\left[\dfrac{\sqrt{2n+2} - \sqrt{2n-2}}{2 + \sqrt{n^2 - 1}}\right] = \sum_{n=1}^{n} \tan^{-1}\left[\dfrac{\sqrt{\dfrac{n+1}{2}} - \sqrt{\dfrac{n-1}{2}}}{1 + \sqrt{\dfrac{n+1}{2}}, \sqrt{\dfrac{n-1}{2}}}\right]$ (divide by 2)

$= \sum_{n=1}^{n}\left(\tan^{-1}\sqrt{\dfrac{n+1}{2}} - \tan^{-1}\sqrt{\dfrac{n-1}{2}}\right) = \tan^{-1}\left(\sqrt{\dfrac{n+1}{2}}\right) + \tan^{-1}\sqrt{\dfrac{n-1}{2}} - \tan^{-1}\dfrac{1}{\sqrt{2}}$

$\lim_{n\to\infty} \text{sum} = \pi - \tan^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{2} + \cot^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{2} + \tan^{-1}\left(\sqrt{2}\right)$

$\therefore k = \sqrt{2} \Rightarrow k^2 = 2$.

6. 3

Sol. $\dfrac{\dfrac{x}{2}+\dfrac{x}{2}+\dfrac{y}{3}+\dfrac{y}{3}+\dfrac{y}{3}+\dfrac{z}{4}+\dfrac{z}{4}+\dfrac{z}{4}+\dfrac{z}{4}}{9} \geq \dfrac{9}{\dfrac{2}{x}+\dfrac{2}{x}+\dfrac{3}{y}+\dfrac{3}{y}+\dfrac{3}{y}+\dfrac{4}{z}+\dfrac{4}{z}+\dfrac{4}{z}+\dfrac{4}{z}}$

7. 4

Sol. $\Delta = \dfrac{1}{2}ab\sin C \Rightarrow \sqrt{3} = \dfrac{1}{2}2b\sin C \Rightarrow \dfrac{\sqrt{3}}{b} = \sin C$

$$2\left[\cos C+\sqrt{3}\sin C\right]=b+c \qquad \ldots\ldots\ldots(1)$$

$$\cos C=\frac{a^2+b^2-c^2}{2ab}=\frac{4+b^2-c^2}{4b}$$

$$\Rightarrow 2\left(\frac{4+b^2-c^2}{4b}\right)+2\sqrt{3}\left(\frac{\sqrt{3}}{b}\right)=b+c$$

$$\Rightarrow \frac{4+b^2-c^2}{2b}+\frac{6}{b}=b+c \Rightarrow 4+b^2-c^2+12=2b^2+2bc$$

$$\Rightarrow b^2+c^2+2bc=16 \Rightarrow (b+c)=4$$

8. 6

Sol. Given, $x^2-4x+1=0 \Rightarrow x$

$$=\frac{4\pm\sqrt{16-4}}{2}=\frac{4\pm 2\sqrt{3}}{2}=2\pm\sqrt{3}$$

$\therefore$ From the figure, I (a, b) lies on line $y=x$,

So b = a

Also, from

ΔIBD, $\frac{r}{BI}=\sin 30^o \Rightarrow BI=2\left[\text{As, } r=1(\text{given})\right]$

$\therefore BI=2=\sqrt{a^2+a^2} \Rightarrow 4=2a^2 \Rightarrow a=\sqrt{2}$

So, $I=\left(\sqrt{2},\sqrt{2}\right)=(a,b)$ (Given)

Hence, $\left(a^2+b^4\right)=\left(\sqrt{2}\right)^2+\left(\sqrt{2}\right)^4=2+4=6$

